

Project Scout

Quick Start

Last updated: May 31, 2026

Upload this at the start of EVERY Claude or ChatGPT session about Scout. For full technical details, use the Scout Master Summary.

1. Who Is Patrick

Patrick Lippy — creator and developer of Scout. **NOT a professional programmer.** Stroke survivor, dyslexic, blind in right eye, type 1 diabetic.

- Explain everything at **screenshot level**. Keep messages clear and not visually overwhelming.
- Always provide **full paste-ready files, one at a time** — or exact CTRL-F / CTRL-R surgical edits. No snippets. No partial files.
- Wife: **Diana** | Son: **Elijah** (age 9, Scout's biggest fan) | Dog: **Nicolas**

Names must **NEVER** be hardcoded in Scout's code. Always use variables.

- Both Claude and ChatGPT are active collaborators. Cross-review between them is welcome.

2. What Scout Is

Scout is a **calm family companion robot** running on a Samsung Galaxy phone mounted in landscape mode as a permanent face display. Scout has animated eyes, speaks, listens, sees via camera, and remembers the family.

- **Package:** com.example.scoutface | **Language:** Kotlin + C++ NDK
- **Primary test device:** Samsung Galaxy A32 (stress-test device — not the benchmark)
- **Dev device:** Samsung Galaxy Fold 7 (wireless via WiFi from Android Studio)
- **Future hardware:** KEYESTUDIO Mini Tank Kit V2 chassis via Bluetooth (opt-in)
- **App:** \$9.99 one-time purchase on Google Play. No subscriptions. Ever.
- **Brains:** TinyLlama 1.1B (offline, always included) + user's own free Gemini key (online, optional)

3. Scout's Core Philosophy

Scout should feel: Calm. Thoughtful. Quietly alive. Emotionally subtle. Occasionally curious.

Scout should NOT feel: Excited. Scripted. Fake. Cartoonish. Hyperactive. Constantly praising.

- Stability > Features
- Presence > Intelligence
- Honest > Fake cheerful
- Local-first > Cloud dependence
- Predictable > Flashy

4. What Is Working Right Now

- Animated face (ScoutFaceView) — custom Canvas animation
- Speech recognition (Android STT) + Text-to-Speech (TTS)
- Camera — face detection, scene labeling (ML Kit)

- Gemini API integration (online brain, user's own key)
- Memory layers: TruthDb, HabitLayer, PeopleDb, JournalDb, ConversationDb
- Intent router — weather, time, greetings, family facts, downloads
- ScoutPresenceDecider — four time-of-day social modes
- **TinyLlama 1.1B offline brain — VERIFIED on BOTH Galaxy A32 and Fold 7** (May 30, 2026)
- Weather via Open-Meteo (no API key needed)
- Teaching system: “My name is Patrick” → stored permanently
- Conversation history trimmed to 2 turns — confirmed in code
- nPredict capped at 64 tokens — prevents rambling
- Scout-specific persona prompt with 5 few-shot examples

5. Known Issues — Do Not Touch Without Discussion

Iris jitter in ScoutFaceView — A32 only (hardware-related). Fold 7 is unaffected. Needs dedicated visual session for minimum-spec devices.

Mouth looping animation — same root cause as iris jitter. Same session.

STT name recognition — 'Scout' mishearing as 'Gal', 'Scott', 'Out'. This is an STT/phonetic matching issue, NOT a TinyLlama problem. Separate session.

TinyLlama identity drift — persona examples partially working. 'Are you my friend?' example needs restoration (2 CTRL-R edits, pending).

Barge-in deliberately disabled — PARKED. Do not attempt to reactivate without a full dedicated session.

■ CURRENT PRIORITY — update this as goals shift

1. Restore 'Are you my friend?' example — 2 CTRL-R edits in MainActivity.kt (tomorrow)
2. ScoutFaceView session — iris jitter (A32), mouth fix, wire in mood system
3. STT phonetic matching for wake word 'Scout' — separate session
4. MainActivity cleanup and shortening — ongoing
5. Offline Brain Phase 3: Google Play Asset Delivery for TinyLlama

7. Working Rules — Always Apply

- Full paste-ready files only, one at a time. No snippets. No partial files.
- Surgical CTRL-F (find) and CTRL-R (find and replace) edits approved for large files — always specify which file tab to click first.
- One safe change at a time. Build and test before the next change.
- Never touch speech, camera, or download systems without explicit discussion.
- Never touch ScoutFaceView casually — it is Scout's visual heart.
- Upload the Scout Master Summary when working on architecture, NDK, or business decisions.
- Patrick is not a professional programmer — screenshot-level explanations always.

8. Key Files

MainActivity.kt	Main app logic, intent handling, offline brain call
ScoutFaceView.kt	Custom face canvas — all visual animation
LlamaEngine.kt	Offline brain JNI wrapper — WORKING
scout_llama_jni.cpp	C++ JNI bridge — compiled into libscout_llama.so
scout_llama_api.h	Self-contained b8946 declarations
GeminiClient.kt	Gemini HTTP wrapper with cooldown discipline
ScoutPromptBuilder.kt	Builds Gemini system instruction
ScoutGeminiManager.kt	Gemini orchestration layer
ScoutWeatherManager.kt	Live weather via Open-Meteo
ScoutPresenceDecider.kt	Social timing layer
ScoutIntentRouter.kt	Intent type routing
OfflinePromptBuilder.kt	TinyLlama ChatML prompt formatter
CMakeLists.txt	NDK build config

This Quick Start is the session-opener. For full architecture, NDK discoveries, business decisions, and technical reference — upload the Scout Master Summary.